

## Supplementary Material

This file contains the following tables:

1. Table S1 - this table provides a summary of the types of anomalies in publicly available datasets. [M Ali: Remove all columns except "Dataset" and "Types of Anomalies"](#)
2. Table S2 - this table provides a summary of the state-of-the-art studies based on RL.
3. Table S3 - this table provides a summary of the state-of-the-art studies based on Gen AI.

Table S1: Publicly available datasets for anomaly detection.

| Dataset           | Year | Format                | F <sup>1</sup> | E <sup>2</sup> | Domain                | Types of anomalies                                                                             | Setup                                                           |
|-------------------|------|-----------------------|----------------|----------------|-----------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| UNSW-NB15         | 2016 | CSV,<br>PCAP          | 49             | ~2,500,000     | Network IDS           | 9 cat.: Fuzzers, Analysis, Backdoor, DoS, Exploits, Generic, Reconnaissance, Shell-code, Worms | Relatively balanced                                             |
| SWaT              | 2016 | CSV                   | 51             | ~500,000       | ICS (water treatment) | Normal and abnormal                                                                            | Network traffic + data from 51 sensors/actuators                |
| MIMIC-III         | 2018 | CSV,<br>SQL           | 47             | 61,532         | Healthcare            | Policy violation                                                                               | Beth Israel Deaconess Medical Center                            |
| CIC-IDS2017       | 2018 | CSV,<br>PCAP          | 85             | ~3,000,000     | Network               | 15 cat.: DoS, PortScan, Web (XSS, SQLi), Heartbleed, etc.                                      | Generated via realistic simulation                              |
| N-BaIoT           | 2018 | CSV                   | 116            | 7,062,606      | IoT botnet            | 2 cat.: Mirai, BASHLITE                                                                        | 9 commercial IoT devices                                        |
| Bot-IoT           | 2019 | CSV,<br>PCAP          | 47             | ~3,000,000     | IoT                   | 4 cat.: DDoS, DoS, Reconnaissance, Information Theft                                           | Realistic IoT env.; five scenarios via Argua and VMs (Node-RED) |
| ECU-IoHT          | 2020 | CSV                   | 8              | 111,207        | Healthcare            | 4 cat.: ARP spoofing, DoS, Smurf, PortScan                                                     | IoHT monitoring devices                                         |
| ToN-IoT           | 2021 | CSV,<br>PCAP,<br>Logs | 44             | 22,000,000     | IoT/IIoT              | 9 cat.: DoS, DDoS, Backdoor, Injection, Password, Ransomware, XSS, MITM                        | Heterogeneous sources; large-scale network                      |
| IoT-23            | 2021 | PCAP                  | 23             | ~688,000       | IoT malware/botnet    | 12 cat.: Okiru, Mirai, Torii, C&C-HeartBeat                                                    | Real IoT device traffic                                         |
| CIC IoT 2023      | 2024 | CSV,<br>PCAP          | 47             | ~46,000,000    | IoT                   | 7 cat.: DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, Mirai                              | Multiple real IoT devices                                       |
| CIC IoT DIAD 2024 | 2024 | CSV<br>(flow-based)   | 121            | ~3,600,000     | IoT                   | 7 cat.: DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, Mirai                              | 105 devices in realistic IoT topology                           |
| CIC IoMT 2024     | 2024 | CSV                   | 39             | 440,315        | IoMT                  | DDoS, DoS, Recon, MQTT, Spoofing                                                               | IoMT testbed (25 real + 15 simulated devices)                   |

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Table S1: Publicly available datasets for anomaly detection.

| Dataset                     | Year | Format                                        | F <sup>1</sup> | E <sup>2</sup> | Domain                               | Types of anomalies                                                                                | Setup                                                                                                                        |
|-----------------------------|------|-----------------------------------------------|----------------|----------------|--------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| DataSense:<br>CIC IIoT 2025 | 2025 | PCAP,<br>Logs<br>(Elastic-<br>search),<br>CSV | ~80            | 1,628,452      | IIoT (Net-<br>work + Sensor<br>Logs) | 7 cat.: Recon, DoS, DDoS, Web, MITM,<br>Brute Force, Malware (Mirai) + Benign; 49<br>attack types | Time-synchronized network<br>(PCAP) + sensor logs;<br>Binary, Multiclass-8, and<br>Multiclass-50 label setups; 40<br>devices |

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<sup>1</sup> Number of features  
<sup>2</sup> Number of examples.

Table S2: Summary of state-of-the-art studies based on RL.

| Ref. | Year | Dataset                          | RL<br>algorithm          | Task / objective                           | Results                          | Strengths                                                  | Limitations                                   |
|------|------|----------------------------------|--------------------------|--------------------------------------------|----------------------------------|------------------------------------------------------------|-----------------------------------------------|
| [38] | 2023 | ToN-IoT (GPS / Modbus / Weather) | DQN                      | Red teaming / attack planning              | Accuracy = 87.8% (Weather split) | Fast convergence; strong on Weather                        | Lower GPS accuracy (79.7%); Modbus weaker     |
| [11] | 2024 | Smart grid simulator (custom)    | DDQN (CNN & GRU)         | Smart grid anomaly detection               | Accuracy = 95.3%                 | Accurate minority attack detection; real-time adaptability | Simulation only                               |
| [39] | 2023 | NSL-KDD                          | Deep SARSA               | Network intrusion detection                | F1 = 84.6%                       | Superior minority-class detection                          | Evaluated on older datasets                   |
| [40] | 2023 | Not specified                    | Actor-Critic             | Adaptive threat detection / response       | Conceptual                       | Supports adversarial learning; real-time policy updates    | No dataset or empirical validation            |
| [41] | 2023 | IPMSRL (maritime OT simulation)  | MAPPO & IPPO             | Industrial / OT defense                    | Success rate = 99.5% (MAPPO)     | Collaborative defense; configurable environment            | Domain-specific; complex simulator setup      |
| [42] | 2023 | IoT-23                           | Recurrent RL (GRU & DQN) | IoT device threat detection                | F1 = 80.6%                       | Captures sequential IoT dynamics                           | High complexity; training overhead            |
| [47] | 2023 | Traffic management (simulated)   | DRL (PPO / DDPG)         | IoT-based traffic & cyber threat response  | Reward +38% over baseline        | Multi-agent coordination; real-time safety decisions       | Environment-specific; no open dataset         |
| [48] | 2023 | Enterprise network (simulated)   | DQN                      | Real-time network threat detection         | Accuracy = 93.1%                 | Adaptive learning; dynamic policy updates                  | Limited generalizability to public benchmarks |
| [47] | 2023 | Smart city traffic (simulated)   | DRL (policy gradient)    | Smart city cyber-physical threat detection | Accuracy = 91.4%                 | Coordinated traffic-cyber response                         | Complex simulation; replication difficult     |

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Table S2: Summary of state-of-the-art studies based on RL.

| Ref. | Year | Dataset        | RL<br>algo-<br>rithm    | Task / objec-<br>tive                        | Results                                  | Strengths                             | Limitations                                  |
|------|------|----------------|-------------------------|----------------------------------------------|------------------------------------------|---------------------------------------|----------------------------------------------|
| [49] | 2023 | CyberBattleSim | DQL (reward<br>shaping) | Red team attack<br>simulation & de-<br>fense | Steps-to-goal: 27%<br>faster convergence | Reward-optimized<br>adaptive planning | Limited generalization be-<br>yond emulation |

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Table S3: Summary of state-of-the-art studies on Gen AI for cybersecurity.

| Ref. | Gen AI alg.        | Use case / domain                                 | Dataset                                      | Contribution                                                                                          | Limitations / challenges                                                                   |
|------|--------------------|---------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| [9]  | GANs               | Intrusion detection systems                       | CICIDS-2017; NSL-KDD; DARPA; CTU-13; RockYou | Taxonomy of GAN-based IDS approaches; highlights cGAN, DCGAN, WGAN for realistic intrusion patterns   | Relies on legacy datasets (e.g., NSL-KDD); no real-world deployment or adversarial testing |
| [66] | GANs               | Synthetic threat data generation                  | Simulated datasets                           | Uses GANs to generate synthetic threats for IDS; supports adversarial scenario emulation for training | Not validated on high-variance attack sets; no operational deployment                      |
| [72] | GANs               | Network traffic anomaly detection                 | Not specified                                | Applies GAN-based models combining feature extraction and generation                                  | No comparison to state-of-the-art DL; lacks adversarial defense integration                |
| [21] | GANs               | Survey on GANs for IDS                            | CICIDS-2017; NSL-KDD                         | Comprehensive survey; categorizes GAN types by threat model and dataset                               | Uses some outdated datasets; limited coverage of newer GAN variants or multimodal threats  |
| [73] | GANs               | IDS evaluation                                    | NSL-KDD; CICIDS-2017                         | Evaluates WGAN, DCGAN, ACGAN with precision/recall                                                    | Dataset limitations; no longitudinal testing; GAN stability issues under-explored          |
| [69] | GANs               | IDS enhancement                                   | CICIDS-2017                                  | Proposes conditional GAN for attack pattern classification                                            | No experimental implementation; minimal GAN configuration details                          |
| [74] | GANs               | Phishing email defense                            | None                                         | Generates phishing emails to harden classifiers via synthetic data                                    | Email set lacks diversity; no comparison to transformer baselines                          |
| [75] | Adaptive GAN       | Real-time threat detection                        | None                                         | Adaptive GAN for evolving threats with minimal retraining                                             | Not validated in dynamic/distributed environments; assumes feature stability               |
| [76] | GANs, VAEs         | Threat simulation (malware, phishing, intrusions) | Synthetic datasets                           | PCA-GAN hybrid for anomaly synthesis to train robust IDS                                              | No real-time validation; limited dataset scope; no adversarial robustness checks           |
| [67] | GANs, transformers | Anomaly detection                                 | CICIDS-2017                                  | Compares GANs vs transformers on intrusion data (detection rate, convergence, scalability)            | Missing transformer tuning and GAN loss curves; reproducibility unclear                    |
| [68] | GANs, Chat-GPT     | Cyber threat analysis                             | None                                         | Framework outlining Gen AI misuse scenarios and attack simulation potential                           | No implementation or experiments                                                           |

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Table S3: Summary of state-of-the-art studies on Gen AI for cybersecurity.

| Ref. | Gen AI alg.   | Use case / domain                    | Dataset                | Contribution                                                                   | Limitations / challenges                                                          |
|------|---------------|--------------------------------------|------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| [77] | GANs, GPT     | GAN vs GPT in cybersecurity          | CICIDS-2017; synthetic | Contrasts GAN and GPT for anomaly detection; reports accuracy/F1               | Narrow dataset scope; simple GPT models; limited external validity                |
| [78] | LLMs          | Prompt injection, jail-breaking      | None                   | Details jailbreak techniques and risks of unrestricted LLM use                 | No quantitative risk model; lacks detection/mitigation methods                    |
| [71] | LLMs          | Defense simulation                   | None                   | Simulation architecture for cyber training using LLM outputs                   | Not evaluated; no red/blue team testing                                           |
| [70] | LLMs, VAEs    | Adaptive threat generation           | None                   | LLM-driven adaptive response; feedback-based defense orchestration             | No SIEM/SOAR integration; latency/performance metrics absent                      |
| [79] | ChatGPT       | Explainable AI in cybersecurity      | None                   | XAI-enhanced IDS with ChatGPT explanations; user-facing interpretability       | Explanations not compared to ground truth/human baseline                          |
| [80] | ChatGPT       | Social / ethical risks               | None                   | Taxonomy of Gen AI misuse and threat classes; policy/ethics overview           | Lacks empirical risk evaluation; conceptual focus                                 |
| [81] | ChatGPT       | Explainable IDS                      | CICIDS-2017            | Uses ChatGPT to explain IDS decisions with LLM-based rationales                | No adversarial evaluation; runtime overhead not reported; reliance on closed APIs |
| [82] | ChatGPT, LLMs | Policy and regulation risk           | None                   | Analysis of Gen AI risks; layered defense and audit-based mitigation framework | No implementation/validation                                                      |
| [10] | ChatGPT, LLMs | Automated hacking / defense          | None                   | Conceptual analysis: AI-generated phishing, malware, social engineering        | No threat modelling or mitigation strategies                                      |
| [83] | ChatGPT, Bard | LLM capabilities for offence/defense | None                   | Compares ChatGPT vs Bard for attack vector generation (evasiveness, diversity) | Dataset unspecified; metrics mostly qualitative                                   |

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